



WhiteKnightTwo and SpaceShipTwo

Built by the company established by innovative designer Burt Rutan, carrier aircraft WhiteKnightTwo is built to fly a payload to suborbital levels before releasing it. The twin-fuselage aircraft can carry either the rocket ship SpaceShipTwo or LauncherOne—an expendable rocket designed to convey satellites into space. WhiteKnightTwo is one of the largest carbon composite aviation vehicle ever built, its wingspan an impressive 141 ft (43 m).

A GLIDER ROCKET

Flown to 50,000 ft (15,240 m) under another craft's power, SpaceShipTwo is able to bypass traditional problems of having to generate enormous thrust at ground level to reach space. Once

launched, the rocket motor does not have to burn for long before escaping Earth's atmosphere. The problem of reentry has also been elegantly addressed: the space plane's wings pivot upward or "feather" to a 65-degree angle to create extra drag. Teamed with the craft's light materials, this slows the return and eliminates the need for the heat shields or tiles that are usually needed to cope with the friction. At 70,000 ft (21,336 m) the wings return to their original sweptback position and the rocket ship turns glider for the flight back to the runway.

Making an unusual silhouette against the sky, carrier aircraft WhiteKnightTwo conveys SpaceShipTwo on a test flight.